THE COMPOSITION OF CLOSED SUBSCHEME CYCLE FAMILIES IS THE FAMILY OF THE FIBRED PRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file relative-cycles.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Cycle family associated to a closed subscheme (relative-cycles chapter, Example example-family-associated-closed, canonical tag 0H4S). cited at statement line 2875. Defines the constructions [Z/X/Y]_r and [W/Y/S]_e appearing in the statement.
- Coherence of the composition construction (relative-cycles chapter, Lemma lemma-construction-coherent, canonical tag 0H69). cited at proof lines 2879-2882. Specialized to F = O_Z and G = O_W: the composition of the module families is the module family of the tensor product.

Source reference. Original reference 0H4S: sources/stacks-chow/tags.txt, source lines 21362–21362. Cited at statement line 2875: defines the constructions in the statement.

Source reference. Original lemma 0H69: sources/stacks-chow/tags.txt, source lines 21414–21414. Cited at proof lines 2879-2882: the special case F = O_Z, G = O_W.

Lemma 0.2. *Let $f : X \to Y$ and $Y \to S$ be morphisms of schemes, both locally of finite type. Let $r, e \geq 0$. Let $Z \subset X$ be a closed subscheme of relative dimension $\leq r$ over Y. Let $W \subset Y$ be a closed subscheme of relative dimension $\leq e$ over S. If $\alpha = [Z/X/Y]_r$ and $\beta = [W/Y/S]_e$ (Example 0H4S) and Z is flat over Y, then $\alpha \circ \beta = [Z \times_Y W/X/S]_{r+e}$.*

Proof. This is a special case of Lemma 0H69 if we take $\mathcal{F} = \mathcal{O}_Z$ and $\mathcal{F} = \mathcal{O}_W$. □

Date: stacks-project:relative-cycles.tex:local-snapshot-20261012; GNU FDL 1.2 or later.